

Healthcare Accessibility Intelligence Pipeline

End-to-End Data Engineering Case Study for Wrench.ai Data Engineering Internship

Prepared by: Tej Harish More | M.S. Data Science, Rochester Institute of Technology

Project area: Personal Health Informatics accessibility barriers across consumer health devices and assistive technologies

Metric	Value
Raw Reddit candidate posts processed	5,400+
Final enriched records in attachment dataset	377
Columns in final dataset	37
Date coverage	Feb 2018 to May 2026
Source communities represented	8 in final dataset
Total discussion comments captured	4,143
High or critical severity records	177
Mean accessibility relevance score	6.18

This report is designed as a project attachment for Wrench.ai. It presents the project from a data engineering perspective: source acquisition, API-driven ingestion, transformation logic, data quality checks, enriched analytical outputs, and production-relevant monitoring concepts.

1. Business Problem and Project Context

Personal Health Informatics (PHI) tools such as wearables, glucose monitors, mobile apps, dashboards, and connected health platforms can fail users when health data is difficult to read, navigate, interpret, or access through assistive technologies. This project mines public Reddit discussions to identify accessibility barriers, device-level friction points, workarounds, and severity patterns across PHI-related technologies.

The engineering goal was not only to collect text data, but to turn messy public discussion data into a structured, analysis-ready dataset. The pipeline converts unstructured posts and comments into features such as device/platform, accessibility barrier, assistive technology, impact type, severity label, relevance score, topic category, and research-ready flags.

Why this matters for Wrench.ai

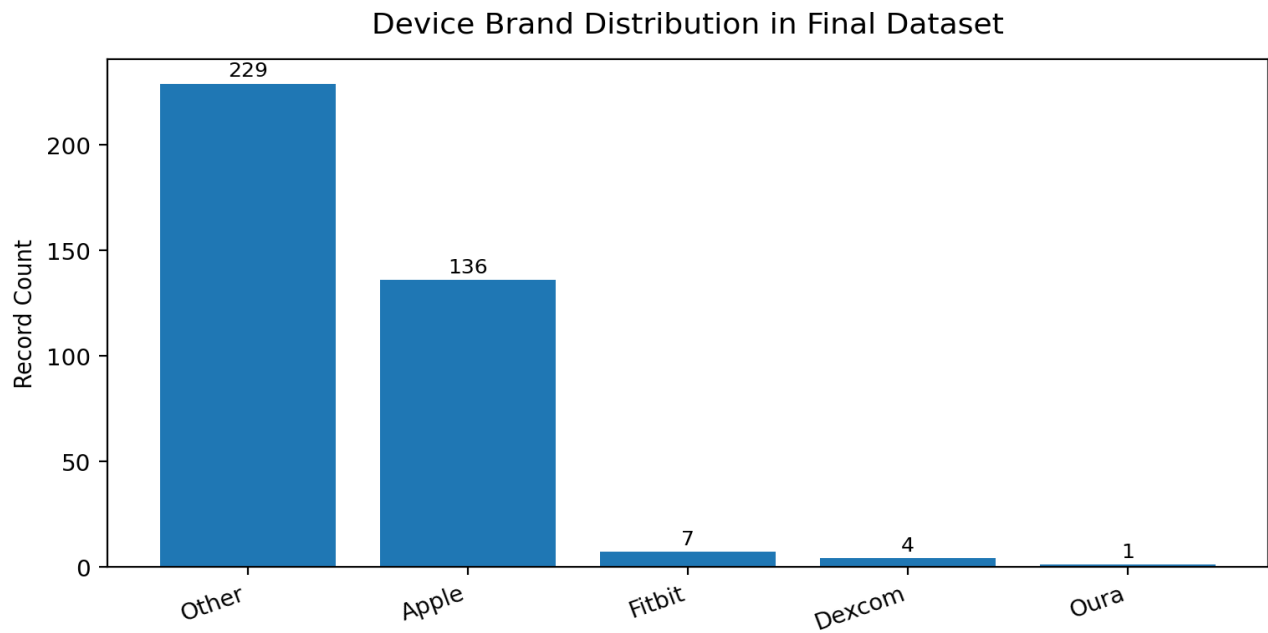
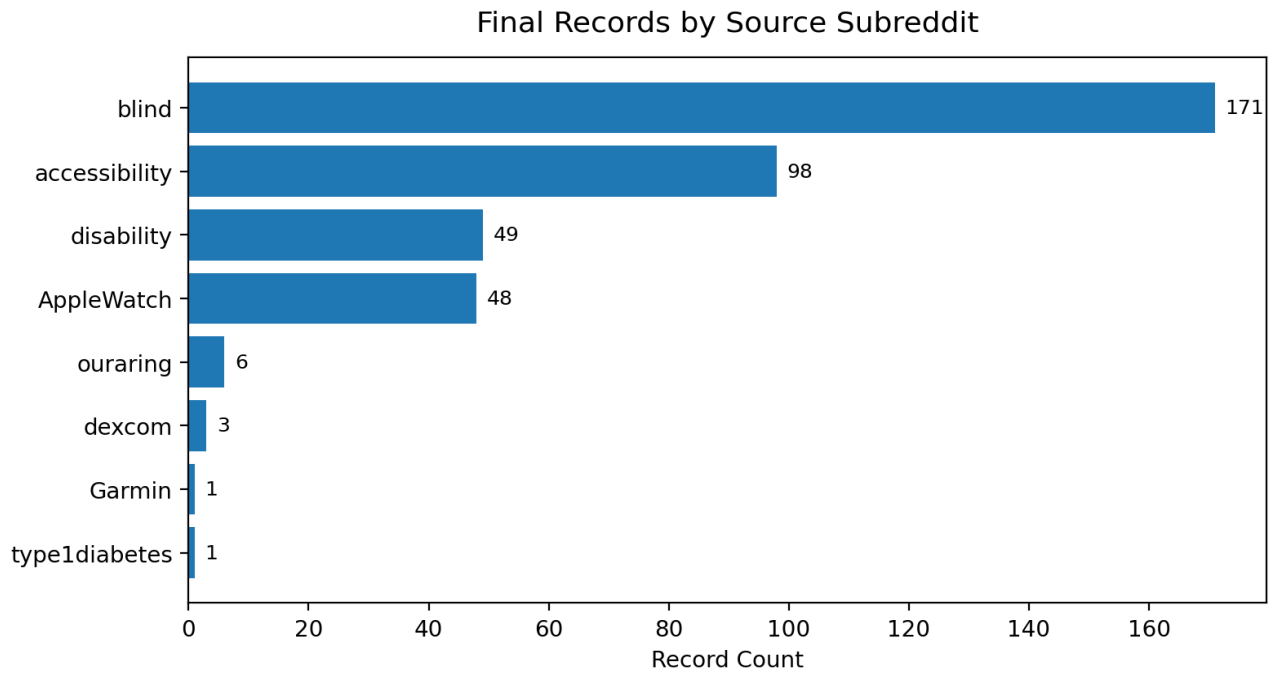
- Wrench.ai converts raw contact, company, and advertising data into actionable lead scores, personas, and personalized outreach. This project demonstrates the same data-engineering pattern: raw multi-source data becomes structured intelligence.
- The work emphasizes pipeline reliability, data quality, metadata enrichment, and downstream AI readiness - all directly aligned with ELT jobs, transformation models, and production data systems.
- The final dataset supports analytics and model training only because the upstream ingestion, cleaning, filtering, and enrichment layers are reliable.

2. Data Sources and Coverage

The source data was collected from the Arctic Shift Reddit archive/API. The pipeline searched public Reddit posts across expert accessibility communities, patient health communities, and consumer technology communities. The goal was to capture both accessibility expertise and real user experiences with health-relevant devices and apps.

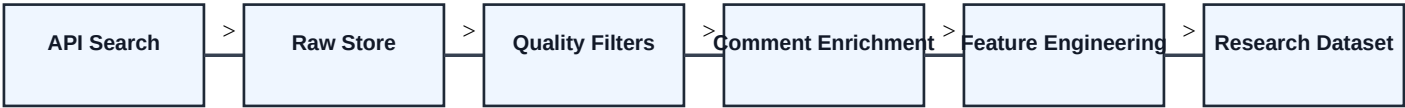
Source group	Candidate communities	Purpose
Accessibility experts	r/blind, r/accessibility, r/disability	Capture assistive technology, screen reader, low vision, and accessibility design discussions.
Patient communities	r/diabetes_t1, r/dexcom, r/type1diabetes	Capture lived experiences with glucose monitoring and diabetes-related PHI devices.
Consumer tech communities	r/AppleWatch, r/Garmin, r/fitbit, r/ouraring	Capture wearable and health-dashboard accessibility experiences.

Candidate devices and platforms included Apple Watch, Apple/iOS and VoiceOver, Android/TalkBack, Dexcom CGM, Fitbit, Garmin, Oura Ring, and general PHI apps or dashboards. The final CSV includes a normalized device_brand field and a more granular device_or_platform field.



3. ETL/ELT Pipeline Architecture

The pipeline was built as a multi-stage workflow that resembles a production ELT system. It first extracts candidate records from a source API, then filters and enriches the records, and finally loads a structured dataset for downstream analytics, visualization, and machine learning.



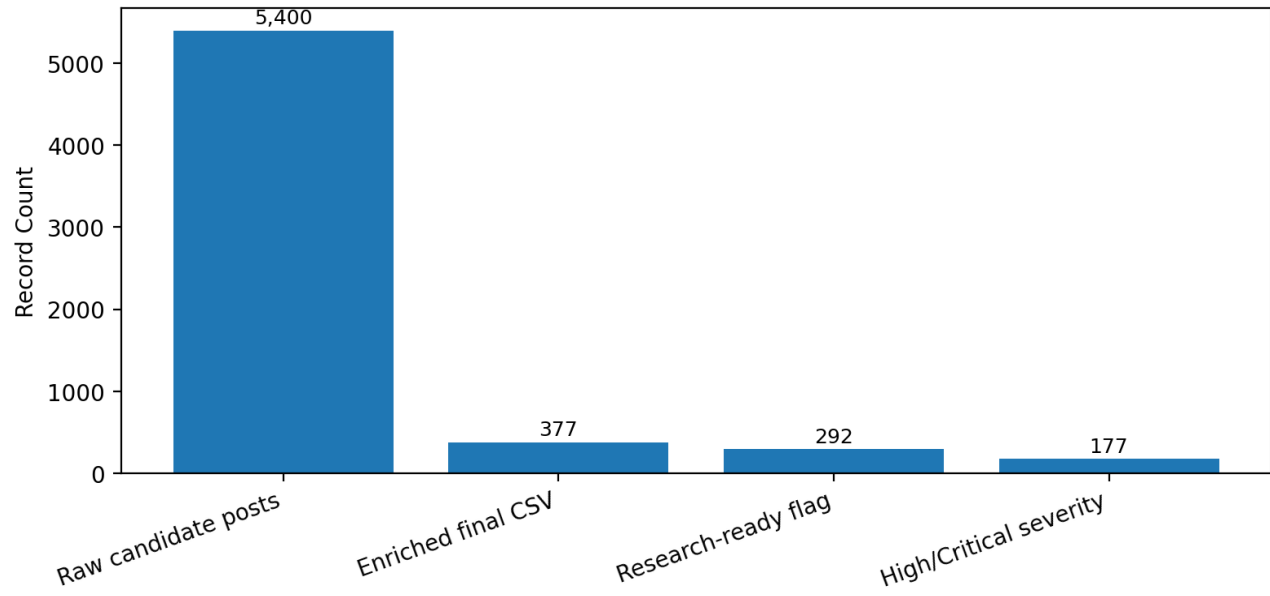
Pipeline stage	Engineering role	Wrench.ai relevance
Extract	Search targeted terms across multiple Reddit communities through the Arctic Shift API.	Similar to integrating external data providers, ad platforms, and enrichment APIs.
Transform	Apply filters, regex checks, accessibility-signal detection, text normalization, and metadata enrichment.	Similar to dbt transformations, data standardization, and ELT model development.
Validate	Remove noise, flag records for manual review, score relevance and severity.	Similar to monitoring data quality before data reaches client-facing intelligence products.
Enrich	Fetch top comments, classify barriers, extract devices/platforms, assign research categories.	Similar to turning raw company/contact/ad data into lead/persona features.
Load	Export a structured 36-column dataset for analytics and downstream AI workflows.	Similar to loading trusted tables into RDS/warehouse layers for product use.

4. Data Quality and Filtering Strategy

The raw data was noisy because public social media archives include short posts, off-topic results, repeated signals, unrelated device discussions, and promotional content. The pipeline used staged filtering to preserve high-value records while reducing manual review effort.

- Comment thresholding removed shallow discussions that were unlikely to contain useful accessibility context.
- Noise filters excluded promotional, off-topic, and irrelevant content.
- Accessibility signal verification retained posts mentioning screen readers, VoiceOver, TalkBack, low vision, unreadable interfaces, contrast, haptics, tremor, dexterity, or related barriers.
- PII review and anonymized author fields supported safer downstream analysis.
- Severity and relevance scoring converted qualitative discussion signals into structured analytical variables.

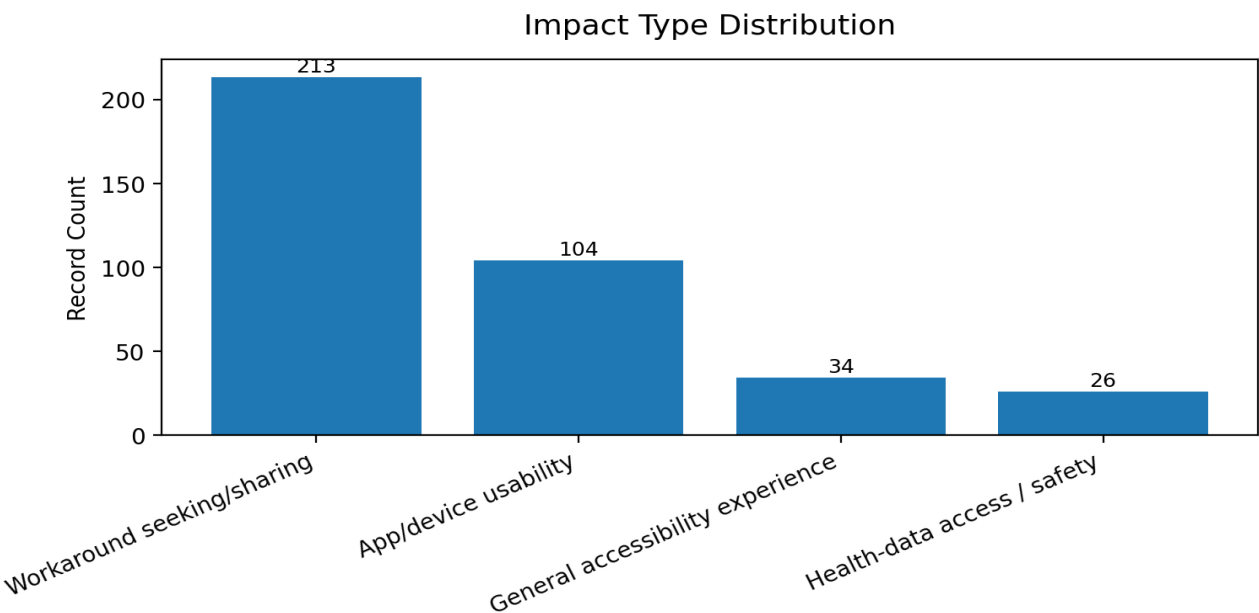
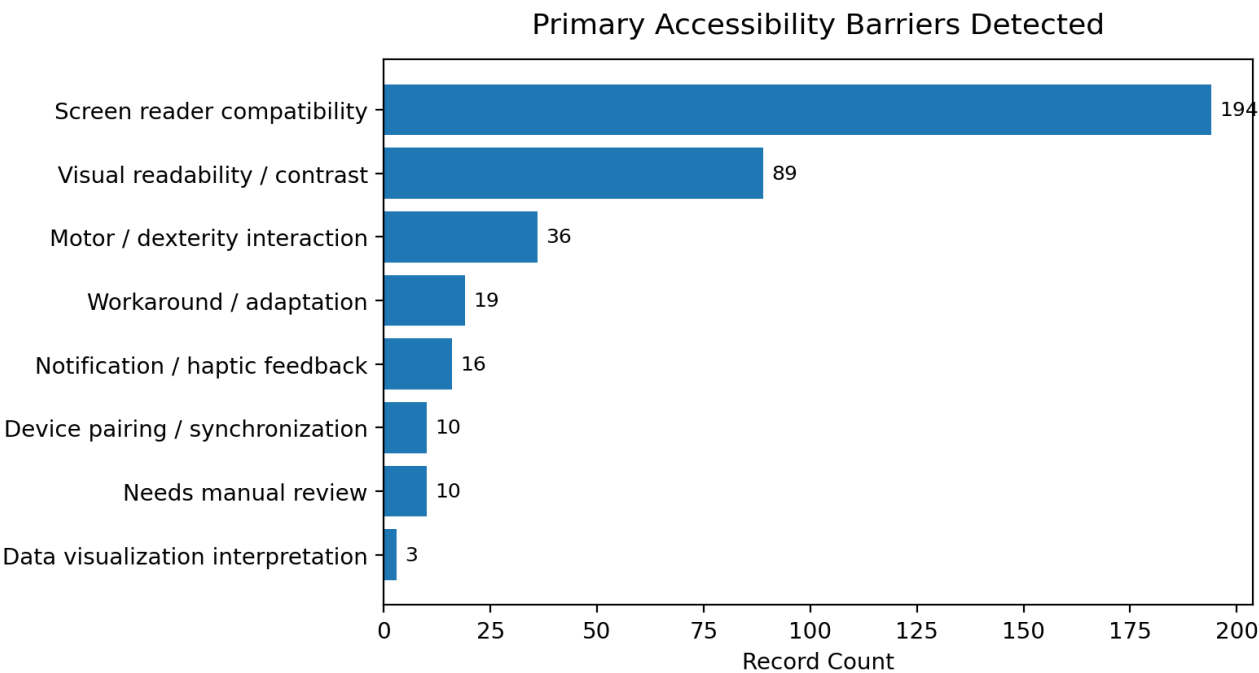
ETL Funnel: From Raw Reddit Signals to Usable Records



5. Enriched Dataset and Analytical Outputs

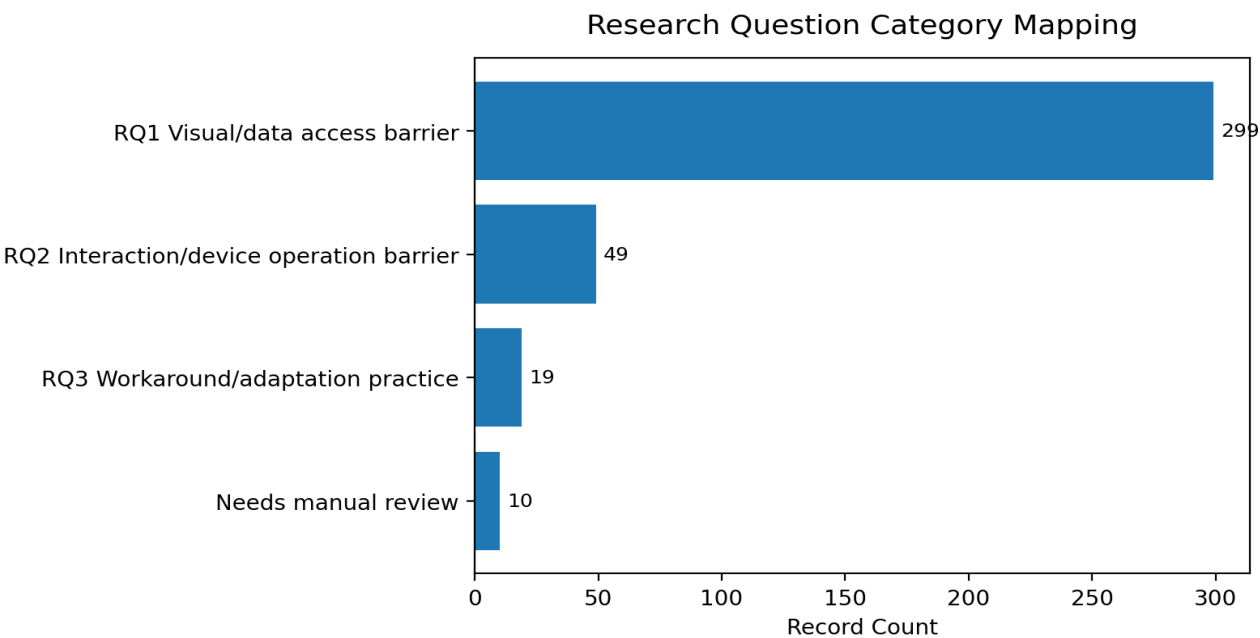
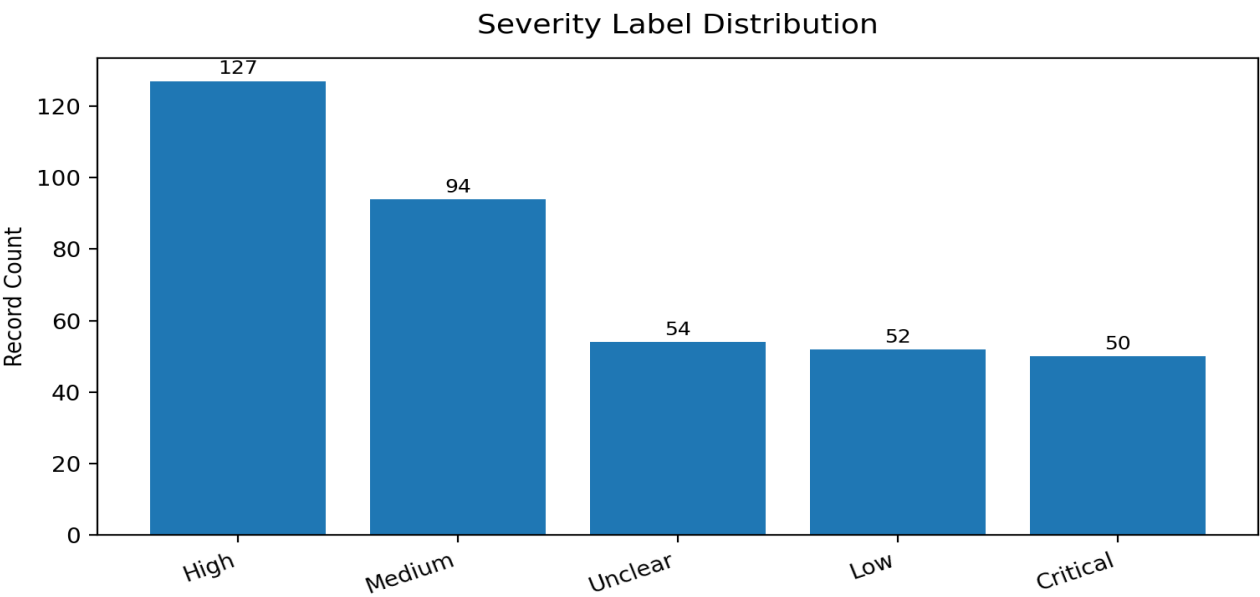
The final dataset contains structured fields that make the Reddit discussions queryable and usable for analytics. Key fields include `source_subreddit`, `subreddit_group`, `device_or_platform`, `device_brand`, `primary_accessibility_barrier`, `accessibility_taxonomy`, `assistive_technology`, `impact_type`, `severity_label`, `severity_score`, `accessibility_relevance_score`, RQ category, topic name, discussion depth score, post engagement metrics, anonymized author, and cleaned text.

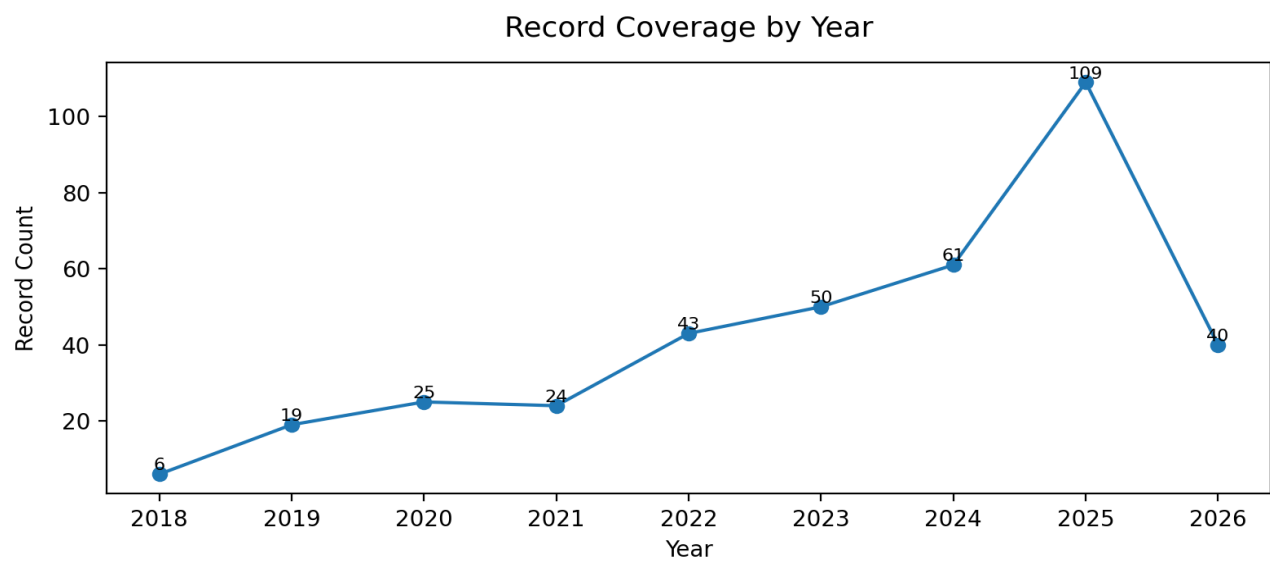
The charts below summarize the most important distributions for a hiring review. They show that the project is not just NLP modeling; it is a complete data product with operational fields, quality signals, and interpretable dimensions.



6. Severity, Research Categories, and Time Coverage

The enriched dataset supports prioritization by severity and research category. This is similar to how production data systems support prioritization and routing: some records become higher-risk, higher-confidence, or higher-value outputs depending on business rules and scoring logic.





7. Production Engineering Translation

Although this project was developed in a research context, the architecture maps naturally to a production data engineering stack. For a Wrench.ai-style environment, the pipeline could be deployed with scheduled extraction, data validation, storage, monitoring, and alerting.

Current project component	Production-equivalent implementation
Python API extraction	AWS Lambda or scheduled container jobs
Intermediate raw records	S3 raw zone or RDS staging tables
Filtering and transformation	dbt models or Python transformation jobs
Data quality checks	Great Expectations, dbt tests, or custom validation rules
Failure monitoring	EventBridge alerts, CloudWatch logs, and Slack/email notifications
Final research-ready CSV	RDS/PostgreSQL analytics tables or warehouse marts
Feature fields for AI analysis	Enriched feature tables for scoring, segmentation, and model input

Recommended monitoring checks

- API response success rate and latency by subreddit/query.
- Daily record counts by source and search term to detect silent data drops.
- Null-rate checks for device, barrier, severity, and research category fields.
- Distribution drift checks for severity labels, barriers, and source communities.
- Deduplication checks using post_id and canonical URL.
- Alerting when extraction volume or final usable records fall below expected thresholds.

8. Fit for the Wrench.ai Data Engineering Internship

Wrench.ai's product depends on reliable data movement from raw sources into lead scores, personas, and personalized outreach intelligence. This project demonstrates my ability to build the upstream layers that make that type of AI product trustworthy.

Wrench.ai requirement	Evidence from this project
Python beyond coursework	Built a functional Python pipeline for API ingestion, filtering, enrichment, and export.
SQL and relational readiness	Final 36-column dataset is normalized enough for PostgreSQL/RDS loading and analytics queries.
API and ETL/ELT curiosity	Used an external Reddit archive API and transformed unstructured data into structured features.
Data quality ownership	Implemented relevance checks, noise filtering, manual review flags, severity scoring, and PII review fields.
Monitoring mindset	Tracked pipeline reduction, final usable records, source distribution, severity distribution, and quality outcomes.
AI/data product alignment	Created clean structured inputs for downstream LLM/NLP analysis, similar to preparing clean inputs for lead scoring and persona systems.
Independent communication	Documented architecture, tradeoffs, metrics, and improvement opportunities in a concise project report.

Conclusion

This project shows that I can work across the data engineering lifecycle: source discovery, API ingestion, cleaning, transformation, enrichment, validation, documentation, and analytics. The work is directly relevant to Wrench.ai because it demonstrates the same core responsibility described in the internship posting: maintaining the data layer between raw external data and the AI systems that depend on it.

Appendix: Dataset Snapshot

Representative columns from the final enriched dataset:

post_id, source_subreddit, subreddit_group, device_or_platform, primary_accessibility_barrier, assistive_technology, impact_type, severity_label, accessibility_relevance_score, research_ready_flag, topic_name_v2, num_comments, score

Field	Distinct values	Most common value
source_subreddit	8	blind (171)
subreddit_group	3	experts (318)
device_brand	5	Other (229)
device_or_platform	18	Other (133)
primary_accessibility_barrier	8	Screen reader compatibility (194)
assistive_technology	30	Not explicit (157)
impact_type	4	Workaround seeking/sharing (213)
severity_label	5	High (127)
RQ_Category_v2	4	RQ1 Visual/data access barrier (299)
topic_name_v2	8	Screen reader compatibility (194)